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COMPUTER IMPLEMENTED METHOD AND SYSTEM FOR SEMANTIC SEGMENTATION OF WORKSITE

Abstract

A semantic segmentation model generates semantic segmentation data indicating locations and types of elements at a worksite. The semantic segmentation model generates the semantic segmentation data based on an image depicting the worksite, elevation data indicated by a digital surface model of the worksite, and slope data derived from the digital surface model. Post-processing may enhance the generated semantic segmentation data based on image processing techniques and contextual information indicated by telematics data associated with operations at the worksite.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to analysis of worksites and, more particularly, to using images and digital surface models to identify elements present at worksites.

BACKGROUND

[0002] Machines, such as haul trucks and other work machines, may perform various activities at worksite, such as a mine site, construction site, or other type of worksite. As an example, a haul truck may be loaded with material at a first location at a worksite, and may transport the material to a second location at the worksite. As another example, an excavator may extract material from an area at the worksite, such that the material may be added to a stockpile or transported to one or more other locations.

[0003] Various elements may be present at worksites, such as machines, equipment, buildings, stockpiles of material, haul roads or other navigable pathways, bodies of water, and/or other elements. The locations, shapes, and/or types of elements at a worksite may change over time. For example, a stockpile of material may be present at the worksite. However, as material is moved away from the stockpile to one or more other locations over a period of time, the boundaries and/or shape of the stockpile, and/or the volume of material in the stockpile, may change.

[0004] Various systems have been developed in the past to identify and/or classify elements at a worksite. For example, Chinese Publication CN112598684A to Luo et al. (hereinafter “Luo”) describes a system that uses semantic segmentation technology classify objects at a mine site based on a color image of the mine site. However, while the system described by Luo may classify objects by evaluating a color image, the system described by Luo may have limited abilities to evaluate other types of data that may increase the accuracy of object classifications generated via semantic segmentation.

[0005] Examples of the present disclosure are directed to overcoming the deficiencies noted above.

SUMMARY

[0006] According to a first aspect of the present disclosure, a computer-implemented method includes receiving, by a computing system including a processor, image data depicting a worksite. The computer-implemented method also includes receiving, by the computing system, a digital surface model (DSM) indicating elevation values corresponding to positions at the worksite. The computer-implemented method additionally includes deriving, by the computing system, slope values associated with the positions based on the elevation values indicated by the DSM. The computer-implemented method further includes generating, by the computing system, and using a semantic segmentation model, semantic segmentation data. The semantic segmentation data. identifies one or more types of elements present at the worksite, and locations of the one or more types of elements at the worksite. The semantic segmentation model is a machine learning model configured to generate the semantic segmentation data based on the image data, the elevation values, and the slope values.

[0007] According to a second aspect of the present disclosure, a computing system includes a processor and a memory. The memory has, stored thereon, computer-executable instructions. The

computer-executable instructions, when executed by the processor, cause the processor to receive, from at least one data source, image data depicting a worksite and a DSM indicating elevation values corresponding to positions at the worksite. The computer-executable instructions also cause the processor to derive slope values associated with the positions, based on the elevation values indicated by the DSM. The computer-executable instructions additionally cause the processor to generate, using a semantic segmentation model, semantic segmentation data. The semantic segmentation data identifies one or more types of elements present at the worksite, and locations of the one or more types of elements at the worksite. The semantic segmentation model is a machine learning model configured to generate the semantic segmentation data based on the image data, the elevation values, and the slope values.
[0008] According to a third aspect of the present disclosure, one or more non-transitory computer-readable media has, stored thereon, computer-executable instructions. The computer-executable instructions, when executed by a processor, cause the processor to receive, from at least one data source, image data depicting a worksite and a DSM indicating elevation values corresponding to positions at the worksite. The computer-executable instructions also cause the processor to derive slope values associated with the positions, based on the elevation values indicated by the DSM. The computer-executable instructions also cause the processor to generate, using a semantic segmentation model, semantic segmentation data. The semantic segmentation data identifies one or more types of elements present at the worksite, and locations of the one or more types of elements at the worksite. The semantic segmentation model is a machine learning model configured to generate the semantic segmentation data based on the image data, the elevation values, and the slope values.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0009] The detailed description is described with reference to the accompanying figures. In the figures, the left-most digit of a reference number identifies the figure in which the reference number first appears. The same reference numbers in different figures indicate similar or identical items.
[0010] FIG. **1** is an exemplary diagrammatic illustration of a worksite segmentation system.
[0011] FIG. **2** is an exemplary diagrammatic illustration of an example of a semantic segmentation model used by the worksite segmentation system.
[0012] FIG. **3** is an exemplary diagrammatic illustration of an example of generation of segmentation data via the semantic segmentation model.
[0013] FIG. **4** is an exemplary diagrammatic illustration of an example of post- processing of segmentation data generated by the semantic segmentation model.
[0014] FIG. **5** is a flowchart illustrating an exemplary process for training the semantic segmentation model.
[0015] FIG. **6** is a flowchart illustrating an exemplary process for using the semantic segmentation model to generate segmentation data associated with a worksite.
[0016] FIG. **7** is a schematic illustration depicting an exemplary architecture of a computing system that executes one or more elements described herein.
[0017] FIG. **8** is an exemplary diagrammatic illustration of an example comparison of segmentation data generated via different techniques.
DETAILED DESCRIPTION
[0018] FIG. **1** is an exemplary diagrammatic illustration of a worksite segmentation system **100**. The worksite segmentation system **100** may include a computing system **102** configured to use a semantic segmentation model **104** to generate segmentation data **106** associated with a worksite

108. The segmentation data **106** may be a prediction that indicates locations, boundaries, and/or types of elements **114** at the worksite **108**.

[0019] For instance, the segmentation data **106** may include one or more image masks associated with one or more corresponding types or classes of elements **114**. Such image masks may identify pixels that represent locations at the worksite **108** at which corresponding types of elements **114** are likely to be present. Accordingly, pixels that the segmentation data **106** identifies as being associated with the same type of element **114** may indicate one or more locations of that type of element **114** at the worksite **108**, as well as shapes and/or boundaries of one or more instances of that type of element **114** at the worksite **108**.

[0020] The worksite **108** may be a mine site, a quarry, a construction site, or any other type of worksite or work environment. Elements **114** at the worksite **108** may include machines **116**, stockpiles of material, haul roads or other navigable paths, loading areas, unloading areas, bodies of water, processing plants, berms, equipment, buildings, and/or other elements.

[0021] The worksite segmentation system **100** may be a computer-implemented system that executes via one or more servers, computers, or other computing devices. Computing elements that execute one or more elements of the worksite segmentation system **100** may be present on-site at the worksite **108**, and/or may be located at one or more back offices or other locations that are remote from the worksite **108**.

[0022] The semantic segmentation model **104** used via the worksite segmentation system **100** may be a machine learning model that is configured to generate the segmentation data **106** based on input data. The input data may include image data **110** that depicts the worksite **108**, a digital surface model (DSM) **112** that indicates elevation values and/or slope values associated with locations at the worksite **108**, and/or other data associated with the worksite **108**.

[0023] The segmentation data **106** may be a prediction that indicates locations, boundaries, and/or types of elements **114** at the worksite **108**. Accordingly, machines **116** may be assigned to perform operations at the worksite **108** based on the locations, boundaries, and/or types of elements **114** indicated by the segmentation data **106**. Similarly, operations and/or productivity at the worksite **108** may be tracked based on the locations, boundaries, and/or types of elements **114** indicated by the segmentation data **106**.

[0024] In some examples, the segmentation data **106** may include, or be displayed as, one or more images representing the worksite **108** that are segmented to identify distinct locations and/or boundaries of one or more types of elements **114** at the worksite **108**. The segmentation data **106** may accordingly include semantic segmentation data, image masks, and/or other data that indicates boundaries of objects, of one or more corresponding classes, that are segmented from an image. For instance, the segmentation data **106** may include a first image showing the locations of one or more stockpiles at the worksite **108**, and a second image showing the locations of one or more roads at the worksite **108**. Examples of image-based segmentation data **106** are shown in FIG. **3**, FIG. **4**, and FIG. **8**, and are discussed further below with respect to those figures.

[0025] One or more machines **116** may operate at the worksite **108**. Such machines **116** may perform one or more types of work operations or tasks at the worksite **108**. A machine **116** may, for example, be a commercial or work machine, such as a mining machine, earth-moving machine, backhoe, scraper, dozer, loader (e.g., large wheel loader, track-type loader, etc.), shovel, truck (e.g., mining truck, haul truck, on-highway truck, off-highway truck, articulated truck, etc.), a crane, a pipe layer, a water truck or other machine that dispenses fluid or other material, farming equipment, or any other type of machine or vehicle. In some examples, multiple machines **116** of the same type, and/or different types, may operate at the worksite **108**.

[0026] The worksite **108** may be a dynamic environment, such that locations, boundaries, and/or types of the elements **114** at the worksite **108** may change over time due to wind, rain, and/or other weather events, operations that are performed at the worksite **108**, and/or other factors. As an example, as machines **116** remove material from a stockpile and transport that material to one or more other locations over a period of time, the shape and/or size of the stockpile at the worksite **108** may change. As another example, if machines **116** are tasked to unload material at a new unloading area at the worksite **108**, a new stockpile of material may be created at the new unloading area. The new stockpile may also grow and/or change shape over time as additional material is added to the stockpile. As yet another example, rainfall may cause puddles or other bodies of water to appear or grow at one or more locations at the worksite **108**. In some situations, such bodies of water may at least temporarily cover haul roads or other elements **114**, for instance until the bodies of water dry up.

[0027] Although the elements **114** at the worksite **108** may change over time, instances of the segmentation data **106** generated via the worksite segmentation system **100** may be predictions that indicate locations, boundaries, and/or types of elements **114** at the worksite **108** at corresponding times. Accordingly, an instance of the segmentation data **106** that is associated with a particular time may indicate the locations, boundaries, and/or types of elements **114** at the worksite **108** at that particular time, without manual identification of the locations, boundaries, and/or types of the elements **114**. Similarly, comparisons of instances of the segmentation data **106** that are associated with different times may be used to identify and/or track changes to elements **114** at the worksite **108** over time.

[0028] As described herein, an instance of segmentation data **106** may be generated by the semantic segmentation model **104** based on image data **110**, a DSM **112** that indicates elevation values and/or slope values, and/or other data. The data used to generate the segmentation data **106** may be captured and/or generated by one or more sensors **118**, such as cameras, Light Detection and Ranging (LiDAR) sensors, and/or other sensors, associated with one or more data sources **120**.

[0029] In some examples, the one or more data sources **120** may include an unmanned aerial vehicle (UAV), such as an autonomous and/or remotely-controlled drone. For instance, as shown in FIG. **1**, an aerial drone may be tasked to fly above the worksite **108** and to use sensors **118** to capture image data **110** and/or a DSM **112** of the worksite **108**. In other examples, the one or more data sources **120** may include other types of aerial sources, such as satellites that pass over the worksite **108**, airplanes or helicopters that fly over the worksite **108**, or other types of vehicles or equipment that may fly or pass over the worksite **108**.

[0030] In still other examples, the one or more data sources **120** may include machines **116** at the worksite **108** that have sensors **118**, and/or may include sensors **118** mounted on buildings, poles, or other fixed locations. In these examples, sensors **118** of such data sources **120** may capture image data **110** depicting at least a portion of the worksite **108**, may capture data that may be used to generate at least a portion of a DSM **112**, and/or may capture data that may be used to enhance or augment data captured by sensors **118** of one or more drones or other aerial data sources **120**.

[0031] The image data **110** may include one or more photographs or other images depicting the worksite **108**. In some examples, the image data **110** may include an RGB image that indicates colors of pixels based on corresponding red values, green values, and blue values. In other examples, the image data **110** may indicate colors using other values and/or other color spaces or profiles. For instance, the image data **110** may include a CMYK image that indicates colors of pixels based on corresponding cyan values, magenta values, yellow values, and black key values.

[0032] The image data **110** may captured as, or may be converted into, an orthophotograph. An orthophotograph may be an adjusted version of a photograph that is modified to present different areas of the worksite **108** at a uniform scale. For example, an orthophotograph may be a version of a captured photograph that has been modified to correct for an angle from which the photograph was captured, to correct for camera lens distortion, to correct for distortions from ground relief, and/or to correct for other issues.

[0033] The DSM **112** may be a topographical model that indicates elevations associated with distinct locations at the worksite **108**. The DSM **112** may have a scale that corresponds to the scale of an orthophotograph or other image data **110**. Accordingly, portions of the DSM **112** that represent areas of the worksite **108** may align with corresponding portions of an orthophotograph or other image data **110** that also represents those areas of the worksite **108**. The DSM **112** and corresponding image data **110** may also represent a state of the worksite **108** at the same point in time, or at similar points in time that are within a threshold period of time from each other.

[0034] The DSM **112** may indicate elevations associated with a ground surface of the worksite **108**, and/or elevations associated with top surfaces of other natural and/or human-made elements **114** present at the worksite **108**. For example, the DSM **112** may be represented as an image that indicates elevation values associated with each pixel of the image. Accordingly, an elevation value associated with a particular pixel of the DSM **112** may indicate an elevation associated with a particular location at the worksite **108** that is represented by that particular pixel, such as an elevation associated with the ground of the worksite **108** at that location or an elevation of the top of another type of element **114** that is present at that location.

[0035] As an example, the DSM **112** may have pixels representing a ground surface of the worksite **108** and other pixels representing a building at the worksite **108**. In this example, the DSM **112** may accordingly indicate that the pixels representing the building are associated with higher elevation values than the pixels representing the ground surface. As another example, the DSM **112** may have pixels representing an uneven ground surface of the worksite **108**. The DSM **112** may accordingly indicate that pixels representing ditches or other depressions in the uneven ground surface have lower elevations than pixels representing hills or other higher portions of the uneven ground surface.

[0036] The DSM **112** may be captured and/or generated based on LiDAR techniques, photogrammetry techniques, and/or other techniques. For example, a LiDAR sensor on a drone or other data source **120** may use lasers to measure distances between the LiDAR sensor and points at the worksite **108**, such that a point cloud of elevation values associated with the worksite **108** may be determined based on the measured distances. As another example, the DSM **112** may be generated via photogrammetry by using a set of photographs of the worksite **108** to generate a three- dimensional model of the worksite **108** that indicates elevation values of distinct points at the worksite **108**.

[0037] Although the DSM **112** may indicate elevation values as discussed above, other information about the worksite **108** may be indicated by the DSM **112** or may be derived from the DSM **112**. For example, a data source **120**, the computing system **102**, and/or another element associated with the worksite segmentation system **100** may derive slope information associated with the worksite **108** based on differences between elevation values indicated by the DSM **112**.

[0038] For instance, the computing system **102** may have a slope derivation component **122** that is configured to derive slope values from elevation values indicated by the DSM **112**. As discussed above, the DSM **112** may indicate elevation values associated with locations at the worksite **108**. The slope derivation component **122** may be configured to derive slope values associated with such locations, based on differences between elevation values indicated by the DSM **112**. As an example, if a first pixel of the DSM **112** that represents a first location at the worksite **108** is associated with first elevation value, and a second pixel of the DSM **112** that represents a second location at the worksite **108** is associated with second elevation value, the slope derivation component **122** may determine a slope between the first location and the second location based on a difference between the first elevation value and the second elevation value.

[0039] In other examples, the DSM **112** may directly indicate slope information. For instance, a data source **120** that captures elevation data and/or generates the DSM **112** may have an instance of the slope derivation component **122**, and may use the slope derivation component **122** to determine slope values that correspond to measured elevation values. The data source **120** may accordingly indicate the slope values in the DSM **112**, or otherwise provide the slope values to the computing system **102** along with elevation values and/or the DSM **112** provided to the computing system **102**.

[0040] As described above, one or more data sources **120** may capture and/or generate image data **110** and/or the DSM **112** associated with the worksite **108**. One or more data sources **120** may also capture and/or generate instances of the image data **110** and/or the DSM **112** at different times. For example, a drone may be tasked to fly above the worksite **108** and capture image data **110** and/or a DSM **112** once per hour, once per day, or an any other regular or irregular schedule.

[0041] The one or more data sources **120** may provide instances of the image data **110** and/or the DSM **112** to the computing system **102** via a network or other data connection. For instance, the one or more data sources **120** may transmit the image data **110** and/or the DSM **112**, and/or corresponding data, to the computing system **102** via a Wi-Fi connection, a cellular data connection, or any other wireless or wired data connection.

[0042] In some examples, the one or more data sources **120** may use sensors **118** to capture data, and may perform processing operations to convert the captured data into image data **110** and/or a DSM **112**. For example, a data source **120** may convert a captured image into an orthophotograph, and/or convert captured distance or elevation data into a DSM **112**, such that the one or more data sources **120** may provide the generated orthophotograph and the DSM **112** to the computing system **102**. The data source **120** may, in some examples, also derive slope data associated with the DSM **112**, and provide the slope data to the computing system **102** in or along with the DSM **112**. In other examples, a data capture source may send captured data to the computing system **102**, such that the computing system **102** may process the captured data to convert a captured image into an orthophotograph, generate a DSM **112**, and/or derive slope data associated with the DSM **112**. For example, a drone may use a LiDAR sensor to measure elevations based on distances between the drone and points at the worksite **108**, and may provide corresponding elevation information to the computing system **102** so that that the computing system **102** may use the elevation information to generate a DSM **112** and/or derive corresponding slope data.

[0043] The semantic segmentation model **104** may use image data **110**, such as an orthophotograph, and elevation and/or slope data indicated by a corresponding DSM **112** to generate segmentation data **106** associated with the worksite **108**. The semantic segmentation model **104** may be a machine learning model, such as a machine learning model based on convolutional neural networks, recurrent neural networks, other types of neural networks, nearest-neighbor algorithms, regression analysis, deep learning algorithms, Gradient Boosted Machines (GBMs), Random Forest algorithms, and/or other types of artificial intelligence or machine learning frameworks.

[0044] The semantic segmentation model **104** may be a computer vision model that is trained and/or configured to determine the shape and/or position of one or more classes of elements **114** within images. As described herein, such images may be associated with input data such as the image data **110**, images indicating elevation values and/or slope values associated with the DSM **112**, and/or other data. As an example, the semantic segmentation model **104** may be an image segmentation model, such as a U-Net model that is based on a deep learning neural network architecture, as discussed further below with respect to FIG. **2**.

[0045] The semantic segmentation model **104** may be trained, via a training system **124**, based on a training data set **126**. In some examples the training system **124** may be executed by the computing system **102**. In other examples, the training system **124** may be executed via a separate computing system that may be different than, and/or remote from, the computing system **102**. For instance, in some examples, the training system **124** may execute via a remote server and may train the semantic segmentation model **104**, and the computing system **102** may access and/or retrieve a trained version of the semantic segmentation model **104** over a network from the training system **124**. The training system **124** may also re-train and/or otherwise update the semantic segmentation model **104** over time, for instance based on new or updated training data sets **126**. Accordingly, the computing system **102** may access and/or use a most recent version of the semantic segmentation model **104** from the training system **124**.

[0046] The training data set **126** may include training image data **128**, training DSMs **130**, and training segmentation data **132**. The training image data **128** may be similar to the image data **110** discussed above, and may include orthophotographs and/or other images depicting one or more example worksites at corresponding times. The training DSMs **130** may be similar to the DSM **112** discussed above, and may indicate elevation data associated with the one or more example worksites in association with corresponding times. The training DSMs **130** may also indicate slope data associated with the one or more example worksites, and/or the training system **124** may use an instance of the slope derivation component **122** to service such slope data from the training DSMs **130**.

[0047] The training segmentation data **132** may indicate locations, boundaries, and/or types of elements **114** at the one or more example worksites at corresponding times. The training segmentation data **132** may be generated by one or more human analysts, such as subject matter experts, who manually evaluated the example worksites in person, and/or based on analysis of images and/or other data associated with the example worksites, to determine the locations, boundaries, and/or types of elements **114** that were present at the example worksites at corresponding times. The training segmentation data **132** may accordingly be considered to be ground truth data indicating the locations, boundaries, and/or types of elements **114** that were present at the example worksites at those times.

[0048] The training data set **126** may accordingly include instances of training image data **128**, training DSMs **130**, and training segmentation data **132** that are associated with the same times and/or the same example worksites. The training system **124** may accordingly use the training data set **126** to train the semantic segmentation model **104** to predict, based on the training image data **128** and the training DSMs **130**, the locations, boundaries, and/or types of elements **114** that the training segmentation data **132** indicates were present at corresponding example worksites at corresponding times.

[0049] As an example, the training system **124** may use a supervised machine learning approach to train the semantic segmentation model **104**. In such a supervised machine learning approach, the training system **124** may use the locations, boundaries, and/or types of elements **114** indicated by the training segmentation data **132** as labels. The training system **124** may also determine which features, and/or which combinations of features, associated with the training image data **128** and the training DSMs **130** may be used to generate predictions of segmentation data that most accurately reproduce the labeled locations, boundaries, and/or types of elements **114** indicated by the training segmentation data **132**. The training system **124** may also determine and/or adjust other parameters of the semantic segmentation model **104** during training of the semantic segmentation model **104**. For example, during training of the semantic segmentation model **104**, the training system **124** may adjust which features are used by the semantic segmentation model **104**, adjust a number of features that are used by the semantic segmentation model **104**, adjust weights associated with the features, and/or determine other parameters that may be used to generate predictions of segmentation data that most accurately reproduce the labeled locations, boundaries, and/or types of elements **114** indicated by the training segmentation data **132**. The training system **124** may accordingly train and/or re-train the semantic segmentation model **104** by selecting different features, selecting different combinations of features, adjusting weights associated with different features, and/or otherwise adjusting parameters of the semantic segmentation model **104** until the semantic segmentation model **104** is able to use the training image data **128** and the training DSMs **130** to generate predictions of segmentation data that match the training segmentation data **132** to at least a threshold degree of accuracy.

[0050] In some examples, the training system **124** may select random patches of the training image data **128** and corresponding training DSMs **130** that represent the same areas of example worksites at the same times, and train the semantic segmentation model **104** based on the selected patches. For example, rather than training the semantic segmentation model **104** based on an entire orthophotograph of an example worksite and corresponding elevation values and/or slope values associated with the entire example worksite indicated by a training DSM **130**, the training system **124** may train the semantic segmentation model **104** based on smaller rectangles that are randomly selected from within the orthophotograph, along with elevation values and/or slope values that correspond with areas of the example worksite depicted by the randomly-selected rectangles. However, in other examples, the training system **124** may training the semantic segmentation model **104** based on an entire orthophotograph of an example worksite and corresponding elevation values and/or slope values associated with the entire example worksite indicated by a training DSM **130**.

[0051] After the training system **124** has trained the semantic segmentation model **104**, the computing system **102** may use a trained instance of the semantic segmentation model **104** to generate segmentation data **106** associated with the worksite **108** based on image data **110** and corresponding DSMs **112** associated with the worksite **108**. For example, the semantic segmentation model **104** may obtain image data **110** and a corresponding DSM **112** that both reflect a state of the worksite **108** at a particular time. In some examples, the semantic segmentation model **104** may also obtain slope data associated with the DSM **112**, for instance from the slope derivation component **122**. The semantic segmentation model **104** may use instances of predictive data features, identified via the training of the semantic segmentation model **104**, that are indicated by image data **110** and data indicated by and/or derived from the DSM **112** to generate segmentation data **106** associated with the particular time. The generated segmentation data **106** may indicate predictions of locations, boundaries, and/or types of elements **114** at the worksite **108** at the particular time.

[0052] The semantic segmentation model **104** may be trained and/or configured to use and/or evaluate features in multiple channels that are indicated by the training image data **128** and/or the training DSMs **130**. Accordingly, the semantic segmentation model **104** may use instances of those features, in multiple channels, that are indicated by image data **110** and corresponding DSMs **112** to generate segmentation data **106**.

[0053] As an example, if the training image data **128** and/or the image data **110** includes color images, the semantic segmentation model **104** may be trained and/or configured to use features of images in channels associated with multiple colors. For instance, if the training image data **128** and/or the image data **110** includes RGB orthophotographs, the semantic segmentation model **104** may be trained and/or configured to use features of an orthophotograph in a red channel, a green channel, and a blue channel.

[0054] As another example, the semantic segmentation model **104** may be trained and/or configured to use elevation features indicated by the training DSMs **130** and/or DSMs **112**. As discussed above, DSMs **112** may indicate elevation values associated with locations at the worksite **108**, and the training DSMs **130** may similarly indicate elevation values associated with locations at example worksites. Accordingly, the features used by the semantic segmentation model **104** may include elevation features, associated with an elevation channel, that are indicated by a DSM.

[0055] As yet another example, the semantic segmentation model **104** may be trained and/or configured to use slope features indicated by the training DSMs **130** and/or instances of the DSM **112**. As discussed above, DSMs **112** may indicate elevation values associated with locations at the worksite **108**, and the training DSMs **130** may similarly indicate elevation values associated with locations at example worksites. The slope derivation component **122** and/or other elements may be configured to derive slope values associated with such locations, based on differences between corresponding elevation values. Accordingly, the features used by the semantic segmentation model **104** may include slope features, associated with a slope channel, that are indicated by a DSM and/or that may be derived from elevation values indicated by a DSM.

[0056] Accordingly, in some examples, the semantic segmentation model **104** may be trained and/or configured to use features in a red channel, a green channel, a blue channel, an elevation channel, and a slope channel to generate an instance of segmentation data **106**. The features of the red channel, the green channel, and the blue channel may be derived from RBG image data **110**, such as an RBG orthophotograph of the worksite **108**. The features of the elevation channel and the slope channel may be indicated by, and/or may be derived from, a DSM **112** associated with the worksite **108**. In other examples, the semantic segmentation model **104** may be trained and/or configured to use features in a different set of channels to generate segmentation data **106**.

[0057] The segmentation data **106** generated by the semantic segmentation model **104** based on the image data **110** and the DSM **112** may indicate predicted locations, boundaries, and/or types of elements **114** that are or were present at the worksite **108** at a time when the image data **110** and the DSM **112** were captured. As an example, if the image data **110** and the DSM **112** were captured at 9:00 AM on a particular day, the image data **110** and the DSM **112** may represent a state of the worksite **108** at 9:00 AM on that particular day. The segmentation data **106**, generated based on the image data **110** and the DSM **112**, may accordingly be a prediction of the locations, boundaries, and/or types of elements **114** present at the worksite **108** at 9:00 AM on that particular day. Another example of segmentation data **106** generated by the semantic segmentation model **104** is shown in FIG. **3**, and is discussed further below with respect to that figure.

[0058] In some examples, the computing system **102** may have a post-processing component **134** configured to augment, enhance, or otherwise adjust the segmentation data **106** generated by the semantic segmentation model **104**. An example of a modification, by the post-processing component **134**, of segmentation data **106** generated by the semantic segmentation model **104** is shown in FIG. **4**, and is discussed further below with respect to that figure. The post-processing component **134** may be configured to use noise correction techniques, binary closing techniques, and/or other image processing techniques to augment, enhance, or otherwise adjust the segmentation data **106**. The post-processing component **134** may adjust the segmentation data **106** to remove likely errors and/or to otherwise cause the segmentation data **106** to more accurately reflect actual locations, boundaries, and/or types of elements **114** at the worksite **108**.

[0059] As an example, the segmentation data **106** may indicate predicted locations of haul roads at the worksite **108**. However, noise in the segmentation data **106** generated by the semantic segmentation model **104** and/or inaccuracies in the predictions indicated by the segmentation data **106** may lead to the segmentation data **106** depicting a relatively small gap between two identified portions of a haul road. The post-processing component **134** may be configured to determine that the gap depicted between the identified portions of the haul road is likely due to noise and/or prediction inaccuracies, and likely does not reflect an actual gap in the haul road. Accordingly, the post-processing component **134** may adjust the segmentation data **106** to fill in the gap and connect the identified portions of the haul road, such that the modified segmentation data **106** depicts a continuous haul road and omits the gap.

[0060] However, as another example, the segmentation data **106** may indicate predicted locations of haul roads at the worksite **108** and also indicate predicted locations of bodies of water at the worksite **108**. In this example, the segmentation data **106** may depict a gap in a haul road, but the segmentation data **106** may also indicate that water is predicted to be located at the position of the depicted gap in the haul road. For instance, a rainstorm may have caused the haul road to flood at the position of the depicted gap. In this example, the post-processing component **134** may use a prediction of water elements **114** indicated by the segmentation data **106** to determine that there is water on the haul road that may make the haul road non- traversable. The post-processing component **134** may accordingly determine not to adjust the segmentation data **106** to fill in the gap in the haul road, as the gap in this example may be likely to be due to identified water that may prevent machines **116** from traveling along the haul road, instead of noise and/or prediction inaccuracies. In other examples, the post-processing component **134** may instead be configured to adjust the segmentation data **106** in this situation, by filling in the gap and connecting the identified portions of the haul road, because the haul road is likely to exist and be present under the identified water.

[0061] The post-processing component **134** may also, or alternately, be configured to use telematics data **136** associated with one or more machines **116** to adjust the segmentation data **106**. For example, the telematics data **136** may indicate locations of the machines **116**, routes traveled by the machines **116**, speeds at which the machines **116** traveled, identifications of operations performed by the machines **116**, information about payloads carried by the machines **116**, such as weights, material types, fuel and/or other payload information, battery status information associated with the machines **116**, images captured by cameras on-board the machines **116**, sensor data captured by sensors **118** on-board the machines **116**, and/or any other type of information. In some examples the telematics data **136** may be transmitted to the computing system **102** from the machines **116** directly. In other example, a worksite management system or other separate system associated with the machines **116** and/or the worksite **108** may receive and/or track telematics data **136** associated with the machines **116**, and may provide the telematics data **136** to the computing system **102**.

[0062] The post-processing component **134** may use the telematics data **136** to modify the segmentation data **106** generated by the semantic segmentation model **104**. For instance, the post-processing component **134** may use the telematics data **136** to determine contextual information associated with operations at the worksite **108** that are associated with elements **114** identified by the segmentation data **106**. The post-processing component **134** may accordingly add corresponding tags, metadata, and/or other contextual data to the segmentation data **106**.

[0063] As an example, the segmentation data **106** generated by the semantic segmentation model **104** may depict locations and/or boundaries of multiple stockpiles at the worksite **108**, but may not initially indicate what types of materials are or were in those stockpiles. However, the telematics data **136** may indicate that machines **116** have been loaded with a first type of material at or near a location of a first stockpile indicted by the segmentation data **106**, and have been loaded with a second type of material at or near a location of a second stockpile indicted by the segmentation data **106**. For instance, the telematics data **136** may include images of material loaded onto the machines **116** that were captured by cameras on-board the machines **116**, such that image recognition techniques may be used to determine the types of material loaded onto the machines **116** at the different locations. Alternately, the telematics data **136** may indicate weights of payloads carried by the machines that may be indicative of the types of material loaded onto the machines **116** at the different locations, or may indicate any other information that may be indicative of the types of material loaded onto the machines **116** at the different locations. Accordingly, the post-processing component **134** may adjust the segmentation data **106**, based on the telematics data **136**, to indicate that the first stockpile depicted in the segmentation data **106** is likely to be associated with the first type of material and that the second stockpile depicted in the segmentation data **106** is likely to be associated with the second type of material.

[0064] As another example, the segmentation data **106** generated by the semantic segmentation model **104** may depict locations and/or boundaries of multiple stockpiles of material at the worksite **108**. However, the segmentation data **106** may not initially indicate whether individual stockpiles are associated with loading areas at which material from the stockpiles are loaded onto machines **116** and are transported away from the stockpiles, or are associated with unloading areas at which material transported by machines **116** are added to the stockpiles. However, the telematics data **136** may indicate that machines **116** have been tasked to perform loading operations at or near the location of a first stockpile identified by the initial segmentation data **106**, and have been tasked to perform unloading operations at or near the location of a second stockpile identified by the initial segmentation data **106**. Accordingly, the post-processing component **134** may adjust the segmentation data **106**, based on the telematics data **136**, to indicate that the first stockpile depicted in the segmentation data **106** is likely to be associated with a loading area and that the second stockpile depicted in the segmentation data **106** is likely to be associated with an unloading area.

[0065] As yet another example, the segmentation data **106** generated by the semantic segmentation model **104** may identify locations of a haul road and a body of water. Similar to an example discussed above, the identified body of water may intersect the haul road, such that the segmentation data **106** depicts a corresponding gap in the haul road. The initial segmentation data **106** may not indicate whether the identified water is a shallow puddle that would not prevent machines **116** from traversing the haul road or a deeper pool or water that would prevent machines **116** from traversing the haul road. However, the telematics data **136** may indicate whether or not machines **116** have been traveling along the haul road and have been passing through the identified water on the haul road. Accordingly, if the telematics data **136** indicates that the identified water has not been preventing machines **116** from traveling along the haul road, the post-processing component **134** may adjust the segmentation data **106** to remove the gap and connect the other identified portions of the haul road as discussed above. However, if the telematics data **136** instead indicates that the identified water has not preventing machines **116** from traveling along the haul road, the post-processing component **134** may determine not to adjust the segmentation data **106** such that the segmentation data **106** depicts the non-navigable gap in the haul road.

[0066] Overall, the semantic segmentation model **104** may use image data **110** and the DSM **112**, for instance based on elevation data and/or slope data indicated by the DSM **112**, to generate segmentation data **106** that indicates locations, boundaries, and/or types of elements **114** at the worksite **108**. The segmentation data **106** may be displayed to one or more users, may be output to other systems, and/or may be used in other ways. For example, as described further below, a worksite management system executed via the computing system **102** or another computing system may use the segmentation data **106** to assign machines **116** to perform operations at the worksite **108**, to track operations and/or productivity at the worksite **108** based on the locations, boundaries, and/or types of elements **114** indicated by the segmentation data **106**, and/or for other purposes. An example of the semantic segmentation model **104** that may be used to generate the segmentation data **106** is discussed further below with respect to FIG. **2**.

[0067] FIG. **2** is an exemplary diagrammatic illustration of an example of the semantic segmentation model **104**. As discussed above, the semantic segmentation model **104** may be a machine learning model. In the example shown in FIG. **2**, the semantic segmentation model **104** may be a U-Net model **200**.

[0068] The U-Net model **200** may be an image segmentation model that is based on a fully convolutional neural network. The U-Net model **200** may determine an element classification associated with each pixel of an image, and may accordingly generate image masks of pixels that indicate locations, shapes, and/or boundaries of the same types of elements **114**. For example, the U-Net model **200** may determine whether a particular pixel of an input image, such as an orthophotograph or other image data **110**, depicts a stockpile, a machine **116**, a haul road, or any other type of element **114**. The U-Net model **200** may accordingly find all of the pixels in the input image that are likely to depict the same type of element **114**, and may generate corresponding segmentation data **106** as an image mask that identifies the pixels associated with that type of element **114**. The U-Net model **200** may generate different image masks associated with different types of elements **114**, such as a first image mask that identifies pixels representing stockpiles, a second image mask that identifies pixels representing haul roads, a third image mask that identifies pixels representing machines **116**, and/or any other image masks associated with other corresponding types of elements **114**.

[0069] As shown in FIG. **2**, the U-Net model **200** may have a “U-shaped” architecture with a contracting path **202** and an expansive path **204**. The contracting path **202** may be associated with an encoder, while the expansive path **204** may be associated with a decoder. An input patch **206**, such as a tile, rectangle, or other portion of an input image, may be provided to the contracting path **202**. The input patch **206** may have pixel values and/or other data associated with multiple channels, such as a red channel, a green channel, and a red channel associated with an orthophotograph or other image data **110**, and an elevation channel and a slope channel associated with a corresponding DSM **112**. The expansive path **204** may output a segmentation data patch **208** that corresponds to the provided input patch **206**. The segmentation data patch **208** may be a portion of segmentation data **106**. Accordingly, the U-Net model **200** may operate separately on different input patches **206** of input data to generate corresponding segmentation data patches **208** of the segmentation data **106**, such that the segmentation data patches **208** may be combined into the full segmentation data **106**.

[0070] The contracting path **202** may include multiple stages that operate to downsample an input patch **206**, and to identify or classify features depicted in the input patch **206**. Each stage of the contracting path **202** may include convolutional layers, and may apply rectified linear unit (ReLU) operations. As an example, first operations **210** such as two 3×3 unpadded convolutions and ReLU operations may be performed at each stage of the contracting path **202**. After each non-final stage of the contracting path **202**, second operations **212** may be performed before the next stage of the contracting path **202**. The second operations **212** may apply max pooling operations, such as 2×2 max pooling. The stages of the contracting path **202** may accordingly double the number of feature channels. A final stage of the contracting path **202** may apply the first operations **210** to generate a feature map. The feature map may be a downsampled version of the input patch **206**.

[0071] The expansive path **204** may then upsample the feature map generated by the contracting path **202**, and may determine information indicating where the features identified via the contracting path **202** are located. The expansive path **204** may also include multiple stages. At each stage of the expansive path **204**, third operations **214** such as upsampling 2×2 up-convolutions may be applied to output of the preceding stage, such as the final stage of the contracting path **202** or a preceding stage of the expansive path **204**, which may halve the number of feature channels.

[0072] Fourth operations **216** involving a residual addition of output from a corresponding stage of the contracting path **202** may also be performed at each stage of the expansive path **204**. For example, the output of the third operations **214** at each stage may be concatenated with a skip layer connection from a corresponding stage of the contracting path **202** via the fourth operations **216**. The skip layers connections used in the fourth operations **216** may allow the expansive path **204** to use feature identification data determined via the stages of the contracting path **202** to determine where the features indicated by the feature map output by the final stage of the contracting path **202** are located within upsampled data generated via the third operations **214**.

[0073] Fifth operations **218**, such as 3×3 convolutions and ReLU operations, may also be applied at each stage of the expansive path **204**. Output of each non-final stage of the expansive path **204** may be provided to the next stage of the expansive path **204**, which may similarly apply the third operations **214**, the fourth operations **216**, and the fifth operations **218**. The final stage of the expansive path **204** may apply sixth operations **220**, such as a 1×1 convolution and/or a sigmoid activation operation, to generate a segmentation data patch **208** that indicates a per-pixel classification of which type of element **114** is likely to be represented by each pixel of the segmentation data patch **208**.

[0074] As discussed above, the U-Net model **200** may operate on different input patches **206** to generate corresponding segmentation data patches **208** of the segmentation data **106**. The input patches **206** and the corresponding segmentation data patches **208** may represent different areas of the worksite **108**. The U-Net model **200** and/or other elements executed by the computing system **102** may combine segmentation data patches **208** representing different areas of the worksite **108** into segmentation data **106** that represents the full worksite **108** or a larger area of the worksite **108**. The segmentation data **106** generated via the U-Net model **200** may indicate locations, boundaries, and/or types of elements **114** at the worksite **108**. Examples of segmentation data **106** are shown in FIG. **3**, FIG. **4**, and FIG. **8**, and are discussed further below with respect to those figures.

[0075] FIG. **3** is an exemplary diagrammatic illustration of an example **300** of generation of segmentation data **106**. As discussed above, the semantic segmentation model **104** may generate segmentation data **106** based on input data that represents a state of the worksite **108**. The input data may include image data **110**, such as an orthophotograph **302** as shown in FIG. **1**. The input data may also include information indicated by, and/or derived from, a DSM **112**, such as elevation data **304** and slope data **306**.

[0076] The orthophotograph **302** may be, or may be derived from, image data **110** captured by a data source **120**. For example, a drone, a satellite, or other data source **120** may use a camera to capture a photograph that depicts the worksite **108**. The data source **120** or the computing system **102** may convert the captured photograph into the orthophotograph **302**, for example to correct for lens distortions, to correct distortions due to an angle at which the photograph was captured, and/or to correct for other issues. Although a grayscale version of the orthophotograph **302** is shown in FIG. **3**, the orthophotograph **302** may be color image, such as an RBG image. Accordingly, pixels of the orthophotograph **302** may indicate data associated with multiple color channels, such as a red channel, a blue channel, and a green channel.

[0077] The elevation data **304** and the slope data **306** may be indicated by, and/or derived from, a DSM **112** captured by a data source **120**. For example, a drone, a satellite, or other data source **120** may use a LiDAR sensor, a camera, and/or other sensors **118** to measure, and/or capture information indicating, elevations of locations at the worksite **108** that are represented by corresponding pixels of the DSM **112**. Such elevation data may be used as the elevation data **304**. Additionally, the data source **120**, the computing system **102**, or another element may derive the slope data **306** from the elevation data **304** indicated by the DSM **112**. For example, the slope derivation component **122** of the computing system **102** may use differences between elevation values associated with different pixels of the DSM **112**, and differences between locations of those pixels within the DSM **112**, to determine corresponding slope values. The determined slope values may be stored or provided as the slope data **306**.

[0078] In some situations, the elevation data **304** and/or the slope data **306** may indicate locations and/or boundaries of elements **114** that may not be directly indicated by color channels of the orthophotograph **302**. For example, if a color of a stockpile of material is relatively similar to the color of a ground surface adjacent to the stockpile, the stockpile may be relatively difficult to detect based on color differences indicated by the orthophotograph **302**. However, the elevation data **304** may indicate that pixels representing the stockpile are associated with higher elevation values than pixels representing the surrounding ground surface. Such differences in the elevation values indicated by the elevation data **304** may accordingly assist the semantic segmentation model **104** in distinguishing pixels that represent the stockpile from pixels that represent the ground surface.

[0079] In some situations, the slope data **306** may also indicate locations and/or boundaries of elements **114** that may not be directly indicated by color channels of the orthophotograph **302** or the elevation data **304**. For example, if a stockpile has a relatively flat top surface, pixels representing the stockpile may be associated with relatively similar elevation values in the elevation data **304** and may be associated with relatively low slope values. However, edges of the stockpile that extend upwards from a relatively flat ground surface to the relatively flat top surface of the stockpile may be associated with relatively high slope values in the slope data **306**. Accordingly, the relatively high slope values in the slope data **306** may assist the semantic segmentation model **104** in identifying pixels representing a boundary that extends around the edges of the stockpile.

[0080] As another example, the roof of a building and the top surface of a stockpile may be relatively flat. The stockpile may have a similar height as the building. Accordingly, pixels representing the building and the stockpile may be associated with similar elevation values in the elevation data **304**. However, the building may have vertical walls, while the stockpile may have more gradually sloped edges that extend upwards from a relatively flat ground surface to the relatively flat top surface of the stockpile. Accordingly, in this example, differences in the slope values associated with the edges of the building and the stockpile may assist the semantic segmentation model **104** in distinguishing pixels that represent the building from pixels that represent the stockpile.

[0081] As shown in FIG. **3**, input data used by the semantic segmentation model **104** may be associated with color channels indicated by the orthophotograph **302**, an elevation channel indicated by the elevation data **304** of the DSM **112**, and/or a slope channel derived indicated by the slope data **306** derived from elevation data **304** of the DSM **112**. The semantic segmentation model **104** may use the input data to generate corresponding segmentation data **106**, such as one or more image masks **308** associated with different types of elements **114**, as shown in FIG. **3**.

[0082] For example, as shown in FIG. **3**, the segmentation data **106** may include an image mask **308** associated with stockpile elements **114** at the worksite **108**. The image mask **308** associated with the stockpile elements **114** may show pixels that are predicted to represent stockpiles in white, and show other pixels that are not predicted to represent stockpiles in black. Although the segmentation data **106** shown in FIG. **3** is an image mask **308** associated with stockpiles, the segmentation data **106** generated by the semantic segmentation model **104** may also include other image masks **308** associated with other types of elements **114**, or the semantic segmentation model **104** may generate different instances of segmentation data **106** that are associated with different types of elements **114**. For instance, in some examples the segmentation data **106** may be, or include, image masks **308** that identify pixels that are predicted to represent roads, pixels that are predicted to represent bodies of water, pixels that are predicted to represent buildings, pixels that are predicted to represent machines **116**, and/or pixels that are predicted to represent any other type or class of element **114**.

[0083] As discussed above, the computing system **102** may have a post-processing component **134** that may modify segmentation data **106** generated by the semantic segmentation model **104**. An example of an initially-generated version of segmentation data **106** being modified by the post-processing component **134** is shown in FIG. **4**, and is discussed further below.

[0084] FIG. **4** is an exemplary diagrammatic illustration of an example **400** of post-processing of segmentation data **106**. The semantic segmentation model **104** may use input data, such as an orthophotograph **302**, elevation data **304**, and/or slope data **306**, to generate segmentation data **106** as discussed above with respect to FIG. **3**. The segmentation data **106** generated and output by the segmentation data **106** may be initial segmentation data **402** as shown in FIG. **4**. In some examples, the initial segmentation data **402** may include noise, have image artifacts, or have other issues that cause the initial segmentation data **402** to inaccurately represent boundaries and/or locations of some elements **114**.

[0085] For example, the initial segmentation data **402** shown in FIG. **4** may be an image mask that identifies pixels predicted to represent haul roads at the worksite **108**. However, as shown in FIG. **4**, the initial segmentation data **402** may have some groups of outlier pixels **404** that do not connect to other groups of pixels that are predicted to represent haul roads. Because haul roads may be designed to be continuous such that machines **116** may travel to any point along the haul roads, it may be relatively likely that such disconnected groups of outlier pixels **404** do not actually represent haul roads. Accordingly, the post-processing component **134** may use noise correction techniques, binary closing techniques, and/or other image processing techniques to adjust the image mask indicated by the initial segmentation data **402** by removing the outlier pixels **404**, in order to generate and/or output modified segmentation data **406** as shown in FIG. **4**. The modified segmentation data **406** may be used as the final segmentation data **106** that is produced and/or output via the computing system **102**.

[0086] In some examples, the post-processing component **134** may use other techniques to modify the initial segmentation data **402** into the modified segmentation data **406**. As an example, the initial segmentation data **402** may include an image mask of haul roads that has continuous areas of pixels that are predicted to represent hauls roads, but that indicates that there is a gap between portions of a haul road. In this example, if the segmentation data **106** also includes an image mask of bodies of water or an image mask of machines **116**, one those image masks indicate that the pixels associated with the gap in the haul road are likely to be associated with water or a machine **116** covering the haul road, the post-processing component **134** may add pixels to the image mask of the haul roads to remove the gap and indicate that the haul road extends continuously through the location where the gap had been.

[0087] In some examples, the post-processing component **134** may use telematics data **136** associated with movements and/or other operations of machines **116** to contextually determine whether or not to remove or add pixels associated with image masks associated with different types of elements **114**. For example, if the initial segmentation data **402** is an image mask of pixels representing haul roads, and includes indicates a group of disconnected outlier pixels **404**, and telematics data **136** indicates that no machines **116** have ever traveled along areas of the worksite **108** represented by the group of disconnected outlier pixels **404**, the post-processing component **134** may determine that the group of disconnected outlier pixels **404** is unlikely to represent an actual haul road and may remove those outlier pixels **404** instead of connecting them to other pixels representing haul roads.

[0088] The post-processing component **134** may also use telematics data **136** to indicate other information in the modified segmentation data **406**. For instance, based on telematics data **136**, the post-processing component **134** may add tags, metadata, and/or other types of contextual data that may provide further information beyond the locations, boundaries, and/or locations of elements **114** at the worksite **108**.

[0089] As an example, the initial segmentation data **402** generated the semantic segmentation model **104** may include an image mask of pixels representing stockpiles, for instance as shown in FIG. **3**. However, although the initial segmentation data **402** may indicate pixels that are likely to represent stockpiles, the initial segmentation data **402** may not indicate which types of material are stored in each of the stockpiles. However, the post-processing component **134** may use telematics data **136** to determine which types of material machines **116** have removed from, or added to, the locations of the stockpiles identified by the initial segmentation data **402**. The post-processing component **134** may add corresponding material classification tags or metadata to the image mask of the stockpiles, to indicate which types of material are likely to associated with individual stockpiles and/or individual pixels associated with corresponding stockpiles. Accordingly, the modified segmentation data **406** output by the post-processing component **134** may be modified to indicate material classifications and/or other types of contextual data, instead of or in addition to being modified to clean up noise or adjust for likely prediction errors.

[0090] As shown in FIG. **3** and FIG. **4**, the semantic segmentation model **104** may use input data, associated with image data **110** and/or a DSM **112**, to generate corresponding segmentation data **106** indicating locations, boundaries, and/or types of elements **114** at the worksite **108**. The post-processing component **134** may modify the segmentation data **106** generated by the semantic segmentation model **104**. Training of the semantic segmentation model **104**, and use of the trained semantic segmentation model **104**, is discussed further below with respect to FIG. **5** and FIG. **6**.

[0091] FIG. **5** is a flowchart **500** illustrating an exemplary process for training the semantic segmentation model **104**. The operations shown in FIG. **5** may be performed by one or more computing systems, such as a computing system that executes the training system **124**. FIG. **7**, discussed further below, describes an example system architecture for such a computing system. In some examples, the computing system that executes the training system **124** and trains the semantic segmentation model **104** may be different, and/or remote from, the computing system **102** that executes a trained instance of the semantic segmentation model **104**. However, in other examples, the same computing system may train the semantic segmentation model **104** and also execute a trained instance of the semantic segmentation model **104**.

[0092] At block **502**, the computing system may obtain the training data set **126**. The training data set **126** may include training image data **128**, such as orthophotographs and/or other images depicting one or more example worksites. The training data set **126** may also include training

DSMs **130**, such as DSMs that indicate elevation values associated with the one or more example worksites. The training data set **126** may also include training segmentation data **132** that indicates locations, boundaries, and/or types of elements **114** that were present at the one or more example worksites. The training segmentation data **132** may be generated by one or more human analysts, such as subject matter experts, and may accordingly be considered to be ground truth data of targets to be predicted during the training of the semantic segmentation model **104**.

[0093] At block **504**, the computing system may determine whether the training data set **126** indicates slope values. The training DSMs **130** may indicate elevation values, from which corresponding slope values may be derived. If slope values have not yet been derived from the elevation values indicated by the training DSMs **130** (Block **504**—No), the computing system may derive such slope values from the elevation values indicated by the training DSMs **130** at block **506**. For example, at block **506**, the computing system may use an instance of the slope derivation component **122** to derive slope values from the training DSMs **130**. The derived slope values may be added to the training data set **126**. After deriving the slope values at block **506**, or if the training data set **126** already indicated slope values (Block **504**—Yes), the computing system may move to block **508**.

[0094] At block **508**, the computing system may divide the training data into input patches. For instance, the computing system may identify sets of training image data **128**, training DSMs **130**, slope values derived from training DSMs **130**, and training segmentation data **132** that represent the states of the example worksites at substantially the same times, and may divide associated pixels and/or corresponding information into input patches that represent the same areas of the example worksites.

[0095] As an example, at block **508** the computing system may divide an orthophotograph of an example worksite into a set of rectangular patches. The computing system may pair the pixels of each rectangular patch of the orthophotograph with elevation values and slope values associated with the locations that are represented by each of the pixels in the rectangular patch. The computing system may also determine which type of element **114** the training segmentation data **132** indicates was present at the locations represented by the pixels of the rectangular patches. The computing system may store the input patches, for instance in local storage associated with the training system **124** and/or in remote storage that is accessible by the training system **124** via a network or other data connection.

[0096] At block **510**, the computing system may select random input patches of training data, from among the patches divided at block **508**. The computing system may access or retrieve the randomly-selected input patches, and may use the randomly-selected input patches to train the semantic segmentation model **104**. In other examples, the computing system may skip block **510**, and may instead train the semantic segmentation model **104** based on all of the input patches of the training data or on other groups of selected input patches.

[0097] At block **512**, the computing system may adjust the semantic segmentation model **104**. For instance, the computing system may select or types and/or numbers of predictive features to be evaluated and/or used to generate segmentation data **106**, selecting or adjusting weights associated with the predictive features, and/or by selecting or adjusting other parameters associated with the semantic segmentation model **104**.

[0098] During a first pass through block **512**, the computing system may select initial parameters for the semantic segmentation model **104**. For example, the computing system may be configured to initialize the semantic segmentation model **104** to evaluate an initial set of features associated with color channels of an orthophotograph **302**, elevation data **304**, and slope data **306** as discussed above with respect to FIG. **3**. As described further below, during subsequent passes through block **512**, the computing system may adjust the parameters of the semantic segmentation model **104**, for instance to increase or decrease weights associated with one or more features, to change the types and/or numbers of features evaluated by the semantic segmentation model **104**, and/or to otherwise change the parameters of the semantic segmentation model **104**.

[0099] At block **514**, the computing system may use the semantic segmentation model **104** to generate segmentation data **106** based on currently-set parameters of the semantic segmentation model **104** and/or features indicated by the training data set **126**. For example, the computing system may execute the semantic segmentation model **104** based on the currently-set parameters to generate segmentation data **106**, for instance based on information indicated by color channels of an orthophotograph **302**, elevation data **304**, and/or slope data **306**. The computing system may execute the semantic segmentation model **104** at block **514** based on one or more input patches of training data, such as one or more input patches that were randomly selected at block **510**.

[0100] At block **516**, the computing system may determine whether an accuracy of the segmentation data **106** generated at block **514**, relative to corresponding training segmentation data **132**, exceeds a threshold. For example, the computing system may compare the segmentation data **106** generated at block **514** against corresponding training segmentation data **132**, to determine how closely the locations, boundaries, and/or types of elements **114** indicated by the generated segmentation data **106** matches the locations, boundaries, and/or types of elements **114** indicated by the training segmentation data **132**. In some examples, the computing system may compare image masks associated with one or more types of elements **114** indicated by the generated segmentation data **106** against corresponding image masks representing the same types of elements **114** indicated by the training segmentation data **132**, to determine whether the image masks have at least a threshold amount or percentage of matching pixels.

[0101] If the accuracy of the generated segmentation data **106**, relative to the training segmentation data **132**, does not exceed an accuracy threshold (Block **516**—No), the computing system may continue to train the semantic segmentation model **104** by returning to block **512** and adjusting the parameters of the semantic segmentation model **104**. The computing system may generate new segmentation data **106** based on the training data and the adjusted parameters at block **514**, and determine at whether the new segmentation data **106** has an accuracy, relative to the training segmentation data **132**, that exceeds the accuracy threshold.

[0102] The computing system may continue training the semantic segmentation model **104** by repeating block **512**, block **514**, and block **516** until the accuracy of segmentation data **106** generated by the semantic segmentation model **104**, relative to the training segmentation data **132**, exceeds the accuracy threshold (Block **516**—Yes). The computing system may accordingly complete the training of the semantic segmentation model **104** at block **518** when the semantic segmentation model **104** is able to generate segmentation data **106** that reproduces corresponding training segmentation data **132** to at least a threshold degree of accuracy.

[0103] When the computing system completes training and/or re-training of the semantic segmentation model **104**, the computing system may store a trained version of the semantic segmentation model **104** in a location, such as server or other data storage location, that is accessible by the computing system **102** that executes the semantic segmentation model **104** to generate segmentation data **106** associated with the worksite **108**, as discussed further below with respect to FIG. **6**.

[0104] In some examples, the computing system may repeat one or more operations shown in FIG. **5** to re-train the semantic segmentation model **104** at one or more times. For example, new and/or updated training image data **128**, training DSMs **130**, and/or training segmentation data **132** may become available over time, such that the computing system may re-train the semantic segmentation model **104** based on a new or updated training data set **126**.

[0105] As another example, over time the computing system may increase the accuracy threshold used at block **516**, and may accordingly re-train the semantic segmentation model **104** to be more accurate relative to the training segmentation data **132**. For example, the training system **124** may train a first version of the semantic segmentation model **104** until the semantic segmentation model **104** is able to reproduce training segmentation data **132** to at least a relatively low threshold degree

of accuracy. Accordingly, the computing system **102** may begin using the first version of the semantic segmentation model **104** after the training system **124** has trained the first version of the semantic segmentation model **104**. However, while the computing system **102** is using the first version of the semantic segmentation model **104**, the training system **124** may increase the accuracy threshold and may train a second version of the semantic segmentation model **104** until the second version of the semantic segmentation model **104** is able to reproduce training segmentation data **132** to a higher threshold degree of accuracy. Accordingly, the computing system **102** may begin using the more accurate second version of the semantic segmentation model **104** after the training system **124** has trained the second version of the semantic segmentation model **104** based on a higher accuracy threshold.
[0106] FIG. **6** is a flowchart **600** illustrating an exemplary process for using the semantic segmentation model **104** to generate segmentation data **106** associated with the worksite **108**. The operations shown in FIG. **6** may be performed by one or more computing systems, such as the computing system **102**. FIG. **7**, discussed further below, describes an example system architecture for such a computing system.
[0107] At block **602**, the computing system may receive image data **110** and a DSM **112** from one or more data sources **120**. For example, a drone or other data source **120** may use a camera to capture image data **110**, such as an orthophotograph. A drone or other data source **120** may also use a LiDAR sensor, a camera, and/or other sensors **118** to measure elevation data indicated by the DSM **112**. The one or more data sources **120** may provide captured data, such as the image data **110** and the DSM **112**, or data from which the image data **110** and the DSM **112** may be derived, to the computing system at block **602** via a network or other data connection.
[0108] At block **604**, the computing system may derive slope values from the DSM **112** received at block **602**. For example, at block **604**, the computing system may use the slope derivation component **122** to derive slope values associated with locations at the worksite **108** based on elevation values corresponding to those locations that are indicated by the DSM **112**.
[0109] At block **606**, the computing system may divide input data into input patches **206**. The input data may include image data **110**, elevation values indicated by the DSM **112**, and slope values derived from the DSM **112**. As an example, the computing system may divide pixels and/or corresponding information associated with an orthophotograph **302**, elevation data **304**, slope data **306** into various input patches that represent the same areas of the worksite **108**. For instance, the computing system may divide the orthophotograph **302** into a set of rectangular patches, and pair the pixels of each rectangular patch with elevation values and slope values associated with the locations that are represented by each of the pixels in the rectangular patch.
[0110] At block **608**, the computing system may select one of the input patches. At block **610**, the computing system may use the semantic segmentation model **104** to generate a semantic segmentation data patch that corresponds to the selected input patch. As discussed above with respect to FIG. **5**, the semantic segmentation model **104** may have been trained to determine which parameters and features in input data can be used to accurately predict corresponding semantic segmentation data **106**. Accordingly, at block **608**, the semantic segmentation model **104** may use those parameters and instances of features indicated by the selected input patch to generate a semantic segmentation data patch that corresponds to the selected input patch. For example, the semantic segmentation model **104** may be trained and/or configured to use features indicted by color channels of the input patch, features of elevation data **304** indicated by the input patch, and features of slope data **306** indicated by the input patch to generate the corresponding semantic segmentation data patch.
[0111] The semantic segmentation data patch generated at block **610** may indicate locations, boundaries, and/or types of the elements **114** present within an area of the worksite **108** represented by the selected input patch and the generated semantic segmentation data patch. In some examples, the generated semantic segmentation data patch may include image masks associated with different

types of elements **114**. Accordingly, an image mask of the semantic segmentation data patch that associated with a particular type of element **114** may identify pixels that represent locations at which that type of element **114** was present at the worksite **108**.

[0112] At block **612**, the computing system may determine if all of the input patches divided at block **606** have been processed via the semantic segmentation model **104** to generate corresponding semantic segmentation data patches. If any input patches have not yet been processed via the semantic segmentation model **104** (Block **612**—No), the computing system may return to block **608** to select another input patch and then use the semantic segmentation model **104** at block **610** to generate a semantic segmentation data patch corresponding to that input patch.

[0113] After semantic segmentation data patches corresponding to all of the input patches have been generated by the semantic segmentation model **104** (Block **614**—Yes), the computing system may combine the semantic segmentation data patches into full segmentation data **106** at block **614**. For example, if the semantic segmentation data patches are rectangular images that each represent a relatively small area of the worksite **108**, the computing system may stitch the semantic segmentation data patches together to generate a larger image that represents a larger area of the worksite **108**. The segmentation data **106** assembled at block **614** may be associated with one or more image masks, such as image masks associated with different types of elements **114** that indicate pixels likely to represent locations at the worksite **108** where those different types of elements were present.

[0114] At block **616**, the computing system may receive telematics data **136** associated with operations of machines **116** at the worksite **108**. The telematics data **136** may indicate locations of the machines **116**, routes traveled by the machines **116**, speeds at which the machines **116** traveled, identifications of operations performed by the machines **116**, information about payloads carried by the machines **116**, such as weights, material types, and/or other payload information, and/or any other type of information. The computing system may receive such telematics data **136** from machines **116** directly, from a worksite management system, or from any other source.

[0115] At block **618**, the computing system may post-process the segmentation data **106** generated at block **614**, for instance to augment, enhance, or otherwise adjust the segmentation data **106**. As an example, the computing system may adjust the segmentation data **106**, using the post-processing component **134**, to remove likely errors and/or to otherwise cause the segmentation data **106** to more accurately reflect actual locations, boundaries, and/or types of elements **114** at the worksite **108**.

[0116] The post-processing component **134** may use noise correction techniques, binary closing techniques, and/or other image processing techniques to augment, enhance, or otherwise adjust the segmentation data **106**. For example, the post- processing component **134** may evaluate image masks of generated segmentation data **106**, and identify outlier pixels **404** that are disconnected from other pixels that represent a particular type of element **114**. The post-processing component **134** may remove such outlier pixels **404** from the image mask based on a determination that the outlier pixels **404** are unlikely to actually represent locations associated with the particular type of element **114**. Alternatively, if the post-processing component **134** determines that the outlier pixels **404** are likely to actually represent locations associated with the particular type of element **114**, the post-processing component **134** may keep the outlier pixels **404** and/or may add other pixels to the image mask to connect the outlier pixels **404** to other pixels that represent locations associated with the particular type of element **114**.

[0117] In some examples, the post-processing component **134** may use the telematics data **136** received at block **616** to augment, enhance, or otherwise adjust the segmentation data **106**. As an example, if segmentation data **106** identifies locations of multiple individual stockpiles, and telematics data **136** indicates what types of material are present at those individual stockpiles, the post-processing component **134** may add tags, metadata, and/or other contextual data to the segmentation data **106** that identifies the types of material associated with the individual stockpiles.

[0118] The post-processing component **134** may also, or alternately, use the telematics data **136** to determine whether to delete or connect outlier pixels **404** in image masks associated with types of elements **114**. For example, the segmentation data **106** may include an image mask associated with a particular type of element **114**, and that image mask may include a set of disconnected outlier pixels **404**. In this example, the post-processing component **134** may use the telematics data **136** to determine whether operations performed by machines **116** the worksite **108** at locations represented by the outlier pixels **404** indicate that the outlier pixels **404** likely do, or do not, actually represent the particular type of element **114**. If the telematics data **136** indicates that operations associated with the particular type of element **114** have been performed at, or proximate to, the locations represented by the particular type of element **114**, the post-processing component **134** may determine to keep and/or connect the outlier pixels **404**. However, if the telematics data **136** indicates that operations associated with the particular type of element **114** have not been performed at, or proximate to, the locations represented by the particular type of element **114**, the post-processing component **134** may instead determine to delete the outlier pixels **404**.
[0119] The segmentation data **106** generated and post-processed via the operations shown in FIG. **6** may be displayed to users of the computing system **102** or other computing systems, may be provided to other systems, and/or may be used in other ways. For example, a worksite management system may use the locations, boundaries, and/or types of elements **114** indicated by the segmentation data **106** to assign machines **116** to perform operations at the worksite **108**, and/or to track operations and/or productivity at the worksite **108**.
[0120] FIG. **7** is a schematic illustration depicting an exemplary architecture of a computing system **700** that executes one or more elements described in the present disclosure. The computing system **700** may include one or more processors **702**, memory **704**, and communication interfaces **706**. The computing system **700** may include one or more computers, servers, or other types of computing devices. Individual computing devices of the computing system **700** may have the system architecture shown in FIG. **7**, or a similar system architecture.
[0121] The computing system **700** may execute one or more elements of the worksite segmentation system **100** described herein, such as the semantic segmentation model **104**, the slope derivation component **122**, the training system **124**, the post-processing component **134**, and/or other elements. The computing system **700** may also, or alternately, execute a worksite management system and/or other elements that receive and/or use segmentation data **106** generated by the semantic segmentation model **104**.
[0122] In some examples, the computing system **700** may include one or more local servers and/or other local computing devices that are physically present at or near the worksite **108**. In other examples, the computing system **700** may include one or more remote servers or other remote computing systems that are located at a remote location relative to the worksite **108**. For instance, the computing system **700** may be executed via one or more remote servers, a cloud computing environment, or other computing systems or elements that are not present at the worksite **108**.
[0123] In some examples, elements associated with the worksite segmentation system **100** may be distributed among, and/or be executed by, multiple computing systems or devices similar to the computing system **700** shown in FIG. **7**. As an example, a trained version of the semantic segmentation model **104** may be executed by a different computing system than a computing system that trains the semantic segmentation model **104**. As another example, the post-processing component **134** may be executed by a different computing system than a computing system that executes the semantic segmentation model **104**. The computing system **700** may, in some examples, include or be part of a cloud computing environment or other distributed system that hosts and/or executes one or more elements associated with the worksite segmentation system **100**.
[0124] The processor(s) **702** of the computing system **700** may operate to perform a variety of functions as set forth herein. The processor(s) **702** may include one or more chips, microprocessors, application specific integrated circuits (ASICs), field programmable gate arrays

(FPGAs) and/or other programmable circuits, central processing units (CPUs), graphics processing units (GPUs), digital signal processors (DSPs), and/or other processing units or components known in the art. In some examples, the processor(s) **702** may have one or more arithmetic logic units (ALUs) that perform arithmetic and logical operations, and/or one or more control units (CUs) that extract instructions and stored content from processor cache memory, and executes such instructions by calling on the ALUs during program execution. The processor(s) **702** may also access content and computer-executable instructions stored in the memory **704**, and execute such computer-executable instructions.

[0125] The memory **704** may be volatile and/or non-volatile computer-readable media including integrated or removable memory devices including random-access memory (RAM), read-only memory (ROM), flash memory, a hard drive or other disk drives, a memory card, optical storage, magnetic storage, and/or any other computer-readable media. The computer-readable media may be non-transitory computer-readable media. The computer-readable media may be configured to store computer-executable instructions that may be executed by the processor(s) **702** to perform the operations described herein.

[0126] For example, the memory **704** may include a drive unit and/or other elements that include machine-readable media. A machine-readable medium may store one or more sets of instructions, such as software or firmware, that embodies any one or more of the methodologies or functions described herein. The instructions may also reside, completely or at least partially, within the processor(s) **702** and/or communication interface(s) **706** during execution thereof by the computing system **700**. For example, the processor(s) **702** may possess local memory, which also may store program modules, program data, and/or one or more operating systems.

[0127] The memory **704** may store data and/or computer-executable instructions associated with elements of the worksite segmentation system **100** described herein. For example, the memory **704** may store data and/or computer-executable instructions associated with the semantic segmentation model **104**, the slope derivation component **122**, the training system **124**, the post-processing component **134**, and/or other elements.

[0128] The memory **704** may also store other modules and data **708** that may be utilized by the computing system **700** to perform or enable performing any action taken by the computing system **700**. For example, the other modules and data **708** may include a platform, operating system, and/or applications, as well as data utilized by the platform, operating system, and/or applications.

[0129] The communication interfaces **706** may include transceivers, modems, interfaces, antennas, and/or other components that may transmit and/or receive data over networks or other data connections. In some examples, the communication interfaces **706** may be wireless communication interfaces and/or wired communication interfaces that the computing system **700** may use to send and/or receive data. As an example, if the computing system **700** executes the semantic segmentation model **104**, the slope derivation component **122**, and/or the post-processing component **134**, the computing system **700** may use the communication interfaces **706** to receive a trained instance of the semantic segmentation model **104** from the training system **124**, to receive image data **110** and/or a DSM **112** from a drone or other data source **120**, to receive telematics data **136** associated with one or more machines **116**, and/or to send generated segmentation data **106** to one or more other elements.

INDUSTRIAL APPLICABILITY

[0130] As described herein, the semantic segmentation model **104** may use image data **110** associated with a worksite **108**, as well as elevation data and slope data indicated by and/or derived from a DSM **112** of the worksite **108**, to generate segmentation data **106** that indicates locations, boundaries, and/or types of elements **114** at the worksite **108**. One or more one or more data sources **120**, such as an aerial drone, may capture the image data **110** and the DSM **112** that is used by the semantic segmentation model **104**. Accordingly, when the one or more data sources **120** captures new image data **110** and a new DSM **112** representing a state of the worksite **108** at a particular time, the semantic segmentation model **104** may generate new segmentation data **106** that indicates locations, boundaries, and/or types of elements **114** present at the worksite **108** at that particular time.

[0131] Data sources **120** may also capture image data **110** and DSMs **112** associated with the worksite **108** at different times, such once per hour, once per day, or based on any other regular or irregular schedule. Accordingly, when new image data **110** and new DSMs **112** associated with the worksite **108** are captured at different times, the semantic segmentation model **104** may generate corresponding instances of the segmentation data **106** that indicate locations, boundaries, and/or types of elements **114** present at the worksite **108** at those different times. Differences between instances of the segmentation data **106** that are associated with different times may accordingly indicate changes over time to the locations, boundaries, and/or types of elements **114** present at the worksite **108**.

[0132] Using the semantic segmentation model **104** to generate segmentation data **106** based on image data **110** and a DSM **112** may allow locations, boundaries, and/or types of elements **114** present at the worksite **108** to be determined automatically in a way that is quicker, cheaper, and/or more efficient than determining such information manually. For example, while human analysts may physically move around the worksite **108** to determine locations, boundaries, and/or types of elements **114** that are present at the worksite **108**, it may take a human a relatively long period of time to survey the entire worksite **108** and to identify all of the elements **114** present at the worksite **108**. Moreover, because the state of the worksite **108** may change relatively rapidly as operations are performed at the worksite **108**, it may be difficult for a human to identify and keep up with dynamic changes to locations, boundaries, and/or types of elements **114** at the worksite **108** over a workday and/or other period of time. However, the semantic segmentation model **104** may automatically generate segmentation data **106** indicating such locations, boundaries, and/or types of elements **114**, and/or changes relative to previous locations, boundaries, and/or types of elements **114** relatively quickly when new image data **110** and new DSMs **112** are captured by one or more drones or other data sources **120**.

[0133] The segmentation data **106** generated automatically by the semantic segmentation model **104** may be used in various ways by one or more systems associated with the worksite **108**. For example, a worksite management system executed via the computing system **102** or another computing system may use the segmentation data **106** to assign machines **116** to perform operations at the worksite **108**, and/or to track operations and/or productivity at the worksite **108**.

[0134] As an example, a worksite management system may assign machines **116** to travel along routes at the worksite **108** that extend along locations of haul roads that are identified by the segmentation data **106**. The worksite management system may also assign machines **116** to travel to, or between, locations of stockpiles, loading areas, unloading areas, or other types of elements **114** that are identified by the segmentation data **106**. The worksite management system may also assign machines **116** to load or unload one or more types of material at such locations based on information indicated by the segmentation data **106**. For instance, if the post-processing component **134** used telematics data **136** to determine types of materials that are likely to be present at different stockpiles, and added corresponding tags or metadata to the segmentation data **106**, the worksite management system may select particular stockpiles that machines **116** should visit based on the locations of those stockpiles indicated by the segmentation data **106** as well as the types of materials that the segmentation data **106** indicates are associated with those stockpiles.

[0135] As another example, the worksite management system may receive telematics data **136** indicating that a particular machine **116** transported a load of material from a first location to a second location at a particular time. In this example the telematics data **136** may have omitted information indicating what type of material was transported by the machine **116** during that particular time. However, the worksite management system may use the segmentation data **106** to determine that the first location was associated with a stockpile of a particular type of material, and thereby determine that the machine **116** transported that particular type of material at the particular time.

[0136] As yet another example, the worksite management system may use segmentation data **106** and/or differences between instances of segmentation data **106** to identify changes to the state of the worksite **108** over time, to track material movement at the worksite **108**, to estimate amounts of material present at various locations at the worksite **108**, and/or for other purposes. For instance, the segmentation data **106** may indicate a location and a boundary of a stockpile of material at a first period of time. The worksite management system may use the boundary of the stockpile indicated by the segmentation data **106** to determine a surface area covered by the stockpile, and may use elevation data and/or slope data indicated by the DSM **112** to determine a height of the stockpile, such that the worksite management system may use the area and the height of the stockpile to estimate a volume of material associated with the stockpile. If subsequent instances of the segmentation data **106** and/or the DSM **112** indicate changes to the area and/or height of the stockpile over a period of time, the worksite management system may use the subsequent instances of the segmentation data **106** and/or the DSM **112** to estimate changes to the volume of material associated with the stockpile and thereby determine how much material has been added to, or removed from, the stockpile.

[0137] The segmentation data **106** generated via the semantic segmentation model **104** described herein may also be more accurate than segmentation data that may be automatically generated by other semantic segmentation techniques. For instance, FIG. **8** is an exemplary diagrammatic illustration of an example comparison **800** of segmentation data generated via different techniques.

[0138] FIG. **8** shows a ground truth representation **802** of actual locations and boundaries of stockpiles at a worksite **108**. FIG. **8** also shows an example of image-based segmentation data **804** that may be automatically generated by a U-Net model based only on an orthophotograph. The image-based segmentation data **804** may be generated by an instance of a U-Net model that has been trained and/or configured to use color data, such as data associated with a red channel, a green channel, and a blue channel, associated with an orthophotograph.

[0139] FIG. **8** also shows an instance of the segmentation data **106** that may be generated by the semantic segmentation model **104** described herein, based on image data **110** in combination with elevation values and slope values indicated by a DSM **112**. Accordingly, although the semantic segmentation model **104** may be trained and/or configured to use color data, such as data associated with a red channel, a green channel, and a blue channel, associated with an orthophotograph or other image data **110**, the semantic segmentation model **104** may be trained and/or configured to use elevation values indicated by a DSM **112** and slope values derived from the DSM **112** in addition to the color data indicated by the image data **110**.

[0140] As shown in FIG. **8**, the image-based segmentation data **804** that a U-Net model may generate from an orthophotograph alone may be noisy. The accuracy of the locations and boundaries of stockpiles indicated by the image-based segmentation data **804**, generated from an orthophotograph alone, may also be low, relative to the actual locations and boundaries of the stockpiles indicated by the ground truth representation **802**.

[0141] However, as shown in FIG. **8**, the segmentation data **106** that the semantic segmentation model **104** generates based on image data **110** in combination with elevation values and slope values indicated by a DSM **112** may be relatively similar to the ground truth representation **802**. Accordingly, the use of elevation values and slope values indicated by a DSM **112**, in addition to image data **110**, may allow the semantic segmentation model **104** to generate segmentation data **106** that is relatively similar to the ground truth representation **802**, and that is thus more accurate at indicting the actual locations and boundaries of the stockpiles than the image-based segmentation data **804** generated from image data **110** alone.

[0142] While aspects of the present disclosure have been particularly shown and described with reference to the embodiments above, it will be understood by those skilled in the art that various additional embodiments may be contemplated by the modification of the disclosed machines, systems, and method without departing from the spirit and scope of what is disclosed. Such embodiments should be understood to fall within the scope of the present disclosure as determined based upon the claims and any equivalents thereof.

Claims

1. A computer-implemented method, comprising: receiving, by a computing system comprising a processor, image data depicting a worksite; receiving, by the computing system, a digital surface model (DSM) indicating elevation values corresponding to positions at the worksite; deriving, by the computing system, slope values associated with the positions based on the elevation values indicated by the DSM; and generating, by the computing system, and using a semantic segmentation model, semantic segmentation data that identifies: one or more types of elements present at the worksite; and locations of the one or more types of elements at the worksite, wherein the semantic segmentation model is a machine learning model configured to generate the semantic segmentation data based on the image data, the elevation values, and the slope values.

2. The computer-implemented method of claim 1, wherein: the semantic segmentation data comprises an image mask associated with a type of element of the one or more types of elements, and pixels of the image mask correspond to particular locations of the type of element at the worksite.

3. The computer-implemented method of claim 2, further comprising post-processing the semantic segmentation data by: identifying, by the computing system, outlier pixels that are disconnected, in the image mask, from other pixels that correspond to the type of element; determining, by the computing system, and based on at least one of image processing techniques or telematics data indicative of operations performed at the worksite, that the outlier pixels are unlikely to represent the type of element associated with the image mask; and deleting, by the computing system, the outlier pixels from the image mask.

4. The computer-implemented method of claim 2, further comprising post- processing the semantic segmentation data by: identifying, by the computing system, outlier pixels that are disconnected, in the image mask, from other pixels that correspond to the type of element; determining, by the computing system, and based on at least one of image processing techniques or telematics data indicative of operations performed at the worksite, that the outlier pixels are likely to represent the type of element associated with the image mask; determining, by the computing system, and based on the at least one of the image processing techniques or the telematics data, that the image mask likely omits additional pixels corresponding to additional locations of the type of element at the worksite; and adding, by the computing system, the additional pixels to connect the outlier pixels with the other pixels in the image mask.

5. The computer-implemented method of claim 1, further comprising post- processing the semantic segmentation data by: receiving, by the computing system, telematics data indicative of operations performed at the worksite; and adding, by the computing system, and based on the telematics data, contextual data to the semantic segmentation data in association with individual instances of the one or more types of elements, wherein the contextual data indicates additional information associated with the individual instances based on the operations performed at the worksite.

6. The computer-implemented method of claim 1, further comprising: providing, by the computing system, the semantic segmentation data to a worksite management system configured to at least one of: assign machines to perform operations at the worksite based on the one or more types of elements, and the locations of the one or more types of elements, identified by the semantic segmentation data, or track productivity at the worksite based on the one or more types of elements, and the locations of the one or more types of elements, identified by the semantic segmentation data.

7. The computer-implemented method of claim 1, wherein the semantic segmentation data is first semantic segmentation data representing a first state of the worksite at a first time, and the method further comprises: receiving, by the computing system, second image data depicting the worksite at a second time; receiving, by the computing system, a second DSM indicating second elevation values corresponding to the positions at the worksite; deriving, by the computing system, second slope values associated with the positions based on the second elevation values indicated by the second DSM; and generating, by the computing system, and using the semantic segmentation model, second semantic segmentation data that identifies: the one or more types of elements present at the worksite at the second time; and locations of the one or more types of elements at the worksite at the second time, wherein differences between the first semantic segmentation data and the second semantic segmentation data are indicative of changes at the worksite between the first time and the second time.
8. The computer-implemented method of claim 1, wherein: the image data is an orthophotograph, and the semantic segmentation model is a U-Net image segmentation model configured to generate the semantic segmentation data based on features in: a plurality of color channels associated with the orthophotograph; an elevation channel corresponding to the elevation values indicated by the DSM; and a slope channel corresponding to the slope values derived from the DSM.
9. The computer-implemented method of claim 1, wherein the image data and the DSM is received from an aerial drone that uses one or more sensors to capture the image data and the DSM while the aerial drone is flying over the worksite.
10. The computer-implemented method of claim 1, wherein: the semantic segmentation model is trained based on a training data set comprising: training image data depicting at least one example worksite; training DSMs indicative of example elevation values and example slope values associated with the at least one example worksite; and training segmentation data that identifies: actual types of elements present at the at least one example worksite; and actual locations of the actual types of elements at the at least one example worksite, and the semantic segmentation model is trained to identify features, indicated by the training image data, the example elevation values, and the example slope values, that allows the semantic segmentation model to predict the actual types of elements and the actual locations of the actual types of elements indicated by the training segmentation data.
11. The computer-implemented method of claim 10, wherein the semantic segmentation model uses instances of the features, identified via training of the semantic segmentation model, and indicated by the image data, the elevation values, and the slope values associated with the worksite, to generate the semantic segmentation data associated with the worksite.
12. A computing system, comprising: a processor; and a memory having stored thereon computer-executable instructions that, when executed by the processor, cause the processor to: receive, from at least one data source: image data depicting a worksite; and a digital surface model (DSM) indicating elevation values corresponding to positions at the worksite; derive slope values associated with the positions, based on the elevation values indicated by the DSM; and generate, using a semantic segmentation model, semantic segmentation data that identifies: one or more types of elements present at the worksite; and locations of the one or more types of elements at the worksite, wherein the semantic segmentation model is a machine learning model configured to generate the semantic segmentation data based on the image data, the elevation values, and the slope values.
13. The computing system of claim 12, wherein: the semantic segmentation data comprises an image mask associated with a type of element of the one or more types of elements, pixels of the image mask correspond to particular locations of the type of element at the worksite, and the computer-executable instructions cause the processor to modify the pixels of the image mask based on at least one of image processing techniques or telematics data indicative of operations performed at the worksite.

14. The computing system of claim 12, wherein: the computer-executable instructions cause the processor to: receive telematics data indicative of operations performed at the worksite; and add, based on the telematics data, contextual data to the semantic segmentation data in association with individual instances of the one or more types of elements, and the contextual data indicates additional information associated with the individual instances based on the operations performed at the worksite.
15. The computing system of claim 12, wherein the computer-executable instructions cause the processor to provide the semantic segmentation data to a worksite management system configured to at least one of: assign machines to perform operations at the worksite based on the one or more types of elements, and the locations of the one or more types of elements, identified by the semantic segmentation data, or track productivity at the worksite based on the one or more types of elements, and the locations of the one or more types of elements, identified by the semantic segmentation data.
16. The computing system of claim 12, wherein: the semantic segmentation model is trained based on a training data set comprising: training image data depicting at least one example worksite; training DSMs indicative of example elevation values and example slope values associated with the at least one example worksite; and training segmentation data that identifies: actual types of elements present at the at least one example worksite; and actual locations of the actual types of elements at the at least one example worksite, and the semantic segmentation model is trained to identify features, indicated by the training image data, the example elevation values, and the example slope values, that allows the semantic segmentation model to predict the actual types of elements and the actual locations of the actual types of elements indicated by the training segmentation data.
17. One or more non-transitory computer-readable media having stored thereon computer-executable instructions that, when executed by a processor, cause the processor to: receive, from at least one data source: image data depicting a worksite; and a digital surface model (DSM) indicating elevation values corresponding to positions at the worksite; derive slope values associated with the positions, based on the elevation values indicated by the DSM; and generate, using a semantic segmentation model, semantic segmentation data that identifies: one or more types of elements present at the worksite; and locations of the one or more types of elements at the worksite, wherein the semantic segmentation model is a machine learning model configured to generate the semantic segmentation data based on the image data, the elevation values, and the slope values.
18. The one or more non-transitory computer-readable media of claim 17, wherein: the semantic segmentation data comprises an image mask associated with a type of element of the one or more types of elements, pixels of the image mask correspond to particular locations of the type of element at the worksite, and the computer-executable instructions cause the processor to modify the pixels of the image mask based on at least one of image processing techniques or telematics data indicative of operations performed at the worksite.
19. The one or more non-transitory computer-readable media of claim 17, wherein: the computer-executable instructions cause the processor to: receive telematics data indicative of operations performed at the worksite; and add, based on the telematics data, contextual data to the semantic segmentation data in association with individual instances of the one or more types of elements, and the contextual data indicates additional information associated with the individual instances based on the operations performed at the worksite.
20. The one or more non-transitory computer-readable media of claim 17, wherein: the semantic segmentation model is trained based on a training data set comprising: training image data depicting at least one example worksite; training DSMs indicative of example elevation values and example slope values associated with the at least one example worksite; and training segmentation data that identifies: actual types of elements present at the at least one example worksite; and actual

locations of the actual types of elements at the at least one example worksite, and the semantic segmentation model is trained to identify features, indicated by the training image data, the example elevation values, and the example slope values, that allows the semantic segmentation model to predict the actual types of elements and the actual locations of the actual types of elements indicated by the training segmentation data.